

RMDL Assessment 2: Data Analysis Report

B282928

1 Background

In English, word frequency is among the most important predictors of lexical recognition reaction times, accounting for substantial variance beyond other word features (Brysbaert et al. 2017). However, word frequency facilitation of lexical recognition is not uniform among speaker populations of a language. Diependaele et al. (2013) found that the word frequency facilitation effect was much steeper in L2 English speakers than native speakers. Diependaele et al describe this asymmetry with the Lexical Entrenchment Account, which predicts that L2 speakers have weaker lexical representations for low-frequency words, while native speakers maintain relatively stronger representations of low-frequency words in the lexicon. Under this account, L2 speakers show longer reaction times on average due to weaker lexical entrenchment. However, for L2 speakers, increases in word frequency lead to proportionally greater facilitation of lexical recognition than for native speakers. This finding has been replicated in lexical decision tasks in Brysbaert et al. (2017), which found that the frequency and language group interaction described above disappeared when controlling for vocabulary size, suggesting that it is lexical entrenchment, and not simply L2 status, that drives the asymmetry.

While the asymmetric relationship between word frequency and word recognition reaction times between L1 and L2 speakers has been studied under the Lexical Entrenchment Account, the theory's application to semantic representations of the word forms is less clear. The Stroop paradigm assesses participants' ability to "inhibit cognitive interference that occurs when the processing of a specific stimulus feature [lexical orthography] impedes the simultaneous processing of a second stimulus attribute [color]," causing a reaction time delay to conflicting orthographic-color pairs as the semantic activation of the written word conflicts with the font color itself (Scarpina and Tagini 2017; Stroop 1935). Therefore, whereas Diependaele et al. (2013) and Brysbaert et al. (2017) investigated lexical recognition, this study seeks to better understand the relationship between word frequency effects and automatic semantic interference between L1 and L2 English speakers through the Stroop task paradigm.

1.1 Research Questions

The Lexical Entrenchment Account predicts lower lexical entrenchment for L2 speakers of English overall, and a greater relative difference in entrenchment between frequent and infrequent English words.

- Does reduced lexical entrenchment for infrequent words in L2 speakers lead to a reduction in automatic semantic activation as seen through the Stroop effect?
- Word frequency has been observed to have an asymmetric impact between L1 and L2 speakers' lexical decision reaction times. Is a similar relationship observed between word frequency and automatic semantic interference in the Stroop effect?

1.2 Research Hypotheses

Following assumptions of the lexical entrenchment account,

1. I predict that L1 speakers of English will have similar lexical entrenchment for infrequent (rare) and frequent (common) word colors. This will lead to credibly longer reaction times between the color and non-color words, demonstrating the Stroop effect, but reaction times between rare and common colors will not be credibly statistically different.
2. On the other hand, L2 English speakers will have much lower lexical entrenchment for rare words in English, leading to reduced automatic semantic interference during the Stroop task. Thus, L2 participants will respond more quickly overall, and will display a credibly greater difference in reaction times between rare and common words due to the greater difference in lexical entrenchment between the two conditions as compared to L1 speakers.

2 Methods

In order to address the main research questions and hypothesis, this study uses a Stroop task paradigm (Stroop 1935). Participants are presented with English color and non-color words of both high and low frequency, using Zipf score from the SUBTLEX UK database as a correlate of frequency. The stimuli set includes 16 words: 4 common colors, 4 rare colors, 4 common non-colors, and 4 rare non-colors. The non-color words are included as controls. A table including the total set of stimuli can be seen in Appendix A. The experiment varied all stimulus words using four font colors: black, blue, green, and brown, matching the common colors in the stimuli set. For the purposes of the Stroop task, common colors were judged as congruent and incongruent based on relation to the font color; all other stimuli types were judged as "control", as the congruence condition is not well defined.

Given that this is a reading task, the current study controls for length of stimulus words in letters by balancing word length across categories. Stimuli were orthogonally matched. Both common and rare conditions contain an identical distribution of word lengths, ranging from three to six letters with a mean length of 4.75 letters for color terms and 4.0 letters for non-color controls.

Stimuli were categorized into the “common” or “rare” condition based on their score on the Zipf scale from 1-7. These values were taken from the SUBTLEX Zipf (UK) data provided by the LexOPS dataset from Taylor et al. (2020). All “common” stimulus items are Zipf scale value 4.8 or above, whereas all “rare” stimulus items are Zipf scale value 3.0 or below.

In total, seven people participated in the experiment. Three were native Standard Mandarin speakers, while four were native English speakers.

At the start of the experiment, participants were presented with the conditions of their participation in the study and asked for informed consent. They were then asked to fill out a language proficiency self-rating form describing their language proficiency in English and in Standard Mandarin with the following labels:

- Native Speaker
- High Proficiency (CEFR C1-C2)
- Intermediate Proficiency (CEFR B1-B2)
- Low Proficiency (CEFR A1-A2)
- No proficiency

An additional freeform text box was included for participants to indicate proficiency in any other languages. Full survey data concerning participant language proficiency self-reporting is shown in Appendix B.

After the language proficiency survey, an information screen was presented describing the experimental task. Participants were instructed that words would appear on the screen, and asked to indicate the font color of the text, ignoring the meaning of the word. They were asked to use the keyboard keys 1, 2, 3, and 4 to indicate the font color of the word, and to answer as accurately and as quickly as possible. Throughout the experiment, a legend displaying the key from number key to color was shown at the bottom of the screen: “Press 1 for BLACK, 2 for BLUE, 3 for GREEN, and 4 for BROWN.”

Following this, eight practice trials were presented to the participant with two congruent, two incongruent, and four control trials. After the practice block, one main trial block was presented to participants. Color word stimuli were presented twice across the four font colors, while non-color word stimuli were presented once across all four font colors, yielding a 2:1 ratio of color to non-color trials and a total of 96 main trials. Reaction time for each trial was measured as the difference between stimulus presentation onset and participant keypress.

2.1 Statistical Analysis

Since this study investigates the difference in the inhibitory Stroop effect when color word and font color are in conflict, all “congruent” trials where the color word matched font color in common words were dropped from the data.

As advocated in Barr et al. (2013), this study employs a maximal random effects structure to analyze experiment data, using a Bayesian linear random effects model from the `brms` package in R to assess the difference in frequency effects between English L1 and L2 participants:

$$\text{ReactionTime} \sim \text{LanguageGroup} * \text{Common_Rare} * \text{Color_Noncolor} + \text{WordLength} + (\text{Common_Rare} + \text{Color_Noncolor} + \text{Length} | \text{Participant})$$

The primary dependent variable is reaction time, modeled using a LogNormal family. This structure explicitly models the interaction between group (L1 vs. L2), frequency of the stimulus word, and type (color word versus non-color word) and stimulus word length on reaction times, while controlling for word length’s impact on reaction times. Given that all participants see multiple levels of word frequency and and type (color versus non-color word), these variables are included as random slopes.

To address the research hypotheses, several data and model parameters are of interest.

1a. L1 (Native) English speakers will have credibly longer reaction times in the common color condition as compared to the common non-color condition, replicating the Stroop effect.

- To evaluate this, I will look at the L1 group reaction times between common color words and common non-color words. If the Stroop effect is replicated, reaction time to name the font color should be credibly lower in the non-color word condition as compared to the color word condition.

1b. L1 (Native) English speakers will **not** have credibly longer reaction times between rare and common items within the color condition.

- To evaluate this, I look at L1 group reaction times between rare and common color words. If the lexical entrenchment account holds, reaction times from L1 English speakers between rare color words and common color words should not be credibly different.

2a. Compared to the L1 group, L2 (Non-native) English speakers will respond to all color items more quickly overall due to lower lexical entrenchment and thus less semantic interference when naming font colors.

- To assess this hypothesis, we will aggregate color word responses across the common and rare condition between L1 English and L2 English speakers. If the hypothesis holds, L2 English speakers will respond more quickly to all color words.

2b. Compared to the L1 group, L2 participants will demonstrate greater relative change in reaction times between rare and common items in the color condition, demonstrating the greater range of lexical entrenchment between the two levels of frequency.

- To look at this part of the hypothesis, we will look at overall reaction times in the color condition for both L1 English and L2 English speakers, comparing between the rare and common conditions. If the hypothesis holds, the range between rare and common condition reaction times should be greater overall in the L2 English group.

2.2 Data Availability Statement

All items relevant to the project are available on GitHub at <https://github.com/owenbraymond/rmdl-B282928>.

Anonymized participant experimental data used for this analysis can be found in the `/results/` folder of this repository.

The stimuli used in the experiment are generated within this file in Appendix A and saved as a .csv in `testable_materials` in this repository.

The Testable experiment file used to run the experiment is generated within this file in Appendix C and saved as a .csv in `testable_materials` in this repository. (This trial file is compatible with Testable as of April 29, 2026.)

3 Data

3.1 Workspace Setup, Package loading

```
library(tidyverse)
library(languageR)
library(ggplot2)
library(devtools)
library(tidybayes)
library(brms)
library(dplyr)
library(plotly)
library(posterior)
library(kableExtra)
pak::pak("JackEdTaylor/LexOPS@*release")
library(LexOPS)
```

3.2 Read and process the data

3.2.1 Importing Trial Data

First, I downloaded all participant experimental result files from Testable as individual .csv files and put them in my repository's "results" folder. I then used the `list.files()` function to create a list of all results files in the `/results/` folder. using `lapply()`, I read all the .csv folders into individual tibbles within `data_lst`, skipping the first 3 rows which contained Testable metadata. I then used `bind_rows()` to combine all trial files into one tibble called `rmdl_data`, and wrote this into my repository's `/data/` folder for use to any other interested researchers who want to start analysis with a combined results file.

```
files_to_read = list.files(
  path = "./results/",
  pattern = "*.csv$",
  recursive = TRUE,
  full.names = TRUE
)
data_lst <- lapply(files_to_read, function(x) {
  read_csv(x, skip = 3) #testable includes 3 rows at the top with metadata,
                        #skip these to get the main trial data
})

rmdl_data <- bind_rows(data_lst, .id = "participant_num") %>%
  filter(condition2 != "congruent" | is.na(condition2))
#Since we are studying the inhibitory effect of word-color pairs,
#we are only going to look at incongruent and control trials.

write_csv(rmdl_data, "data/rmdl_data.csv")
```

3.3 Relevant Columns

`participant_num` : Unique participant ID for each participant.

`stim1` : The word text of the stimulus item.

`color` : The font color of the stimulus item.

`RT` : Participant reaction time to the stimulus item.

`rarity` : Whether the stimulus item belongs to the "rare" or "common" condition.

`color_noncolor` : Whether the stimulus item belongs to the "color" or "non-color" condition.

`Length` : Stimulus item length in letters.

group : Participant group, either “Mandarin” or “English”, representing the participant’s L1.

3.4 Splitting the Data

In the combined experimental data tibble, I used the `type` column to create two sub-tibbles, one for participant language proficiency self-rating information (`survey_responses`), and one for the reaction time data to stimulus items (`rmdl_data_RT`).

```
survey_responses <- rmdl_data %>%
  filter(type == "form") %>%
  mutate(
    english_level = ifelse(head ==
      "What is your English language READING ability?", response, NA),
    mandarin_level = ifelse(head ==
      "What is your Standard Mandarin ( ) language READING ability?",
      response, NA),
    other_level = ifelse(head ==
      "Are you proficient in or learning any other foreign languages?
      (Please include the name of the language
      and your proficiency self-assessment.)"
      , response, NA)
  ) %>%
  group_by(participant_num) %>%
  summarise(
    english_level = first(na.omit(english_level)),
    mandarin_level = first(na.omit(mandarin_level)),
    other_level = first(na.omit(other_level))
  )

rmdl_data <- rmdl_data %>%
  left_join(survey_responses, by = "participant_num")

#creating labels for stats analysis
rmdl_data <- rmdl_data %>%
  mutate(
    group = case_when(
      english_level == "Native Speaker" ~ "English",
      mandarin_level == "Native Speaker" ~ "Mandarin",
      TRUE ~ "Other L2"
    )
  )
```

```

rmdl_data_RT <- rmdl_data %>%
  #We will drop
  filter(type == "test" & condition1 != "practice") %>%
  select(participant_num, type, stim1, color, key, condition1, condition2,
         RT, english_level, mandarin_level, other_level, rarity,
         color_noncolor, Zipf.SUBTLEX_UK, Length, combined_condition,
         congruence_condition, correct, group)

```

3.5 Plotting the data

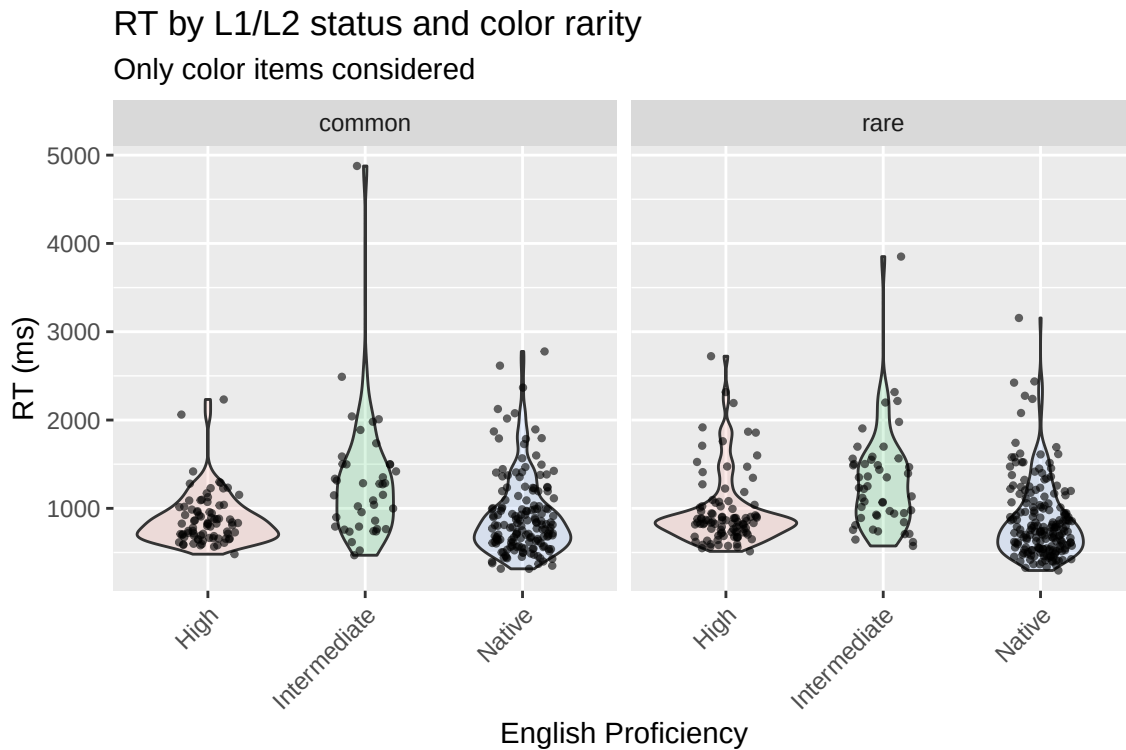
3.5.1 Figure 1.

```

#The code will print the figures,
#but you can find a saved vector graphic version in the 'figures' subfolder.
rawdata_violin <- rmdl_data_RT %>%
  #Shortening X axis labels for readability.
  mutate(
    english_level = case_when(
      english_level == "Native Speaker" ~ "Native",
      english_level == "High Proficiency (CEFR C1-C2)" ~ "High",
      english_level == "Intermediate Proficiency (CEFR B1-B2)" ~
        "Intermediate",
      english_level == "Low Proficiency (CEFR A1-A2)" ~
        "Low",
      TRUE ~ "None"
    )
  ) %>%
  ggplot(aes(x = english_level, y = RT, fill = english_level)) +
  geom_violin(alpha = 0.15) +
  geom_jitter(width = 0.2, alpha = 0.6, size = 0.8) +
  facet_wrap(~rarity) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position="none") +
  labs(
    x = "English Proficiency",
    y = "RT (ms)",
    title = "RT by L1/L2 status and color rarity",
    subtitle="Only color items considered")

rawdata_violin

```



```
ggsave("figures/rawdata_violin.svg")
```

Saving 6 x 4 in image

Figure 1. This figure plots reaction times on the Y axis and groups English proficiency level of participants on the X axis, shown in red, green, and blue for visual clarity. This plot shows that while there was variation between group reaction times, patterns look fairly similar between the common and rare conditions overall.

3.5.2 Figure 2.

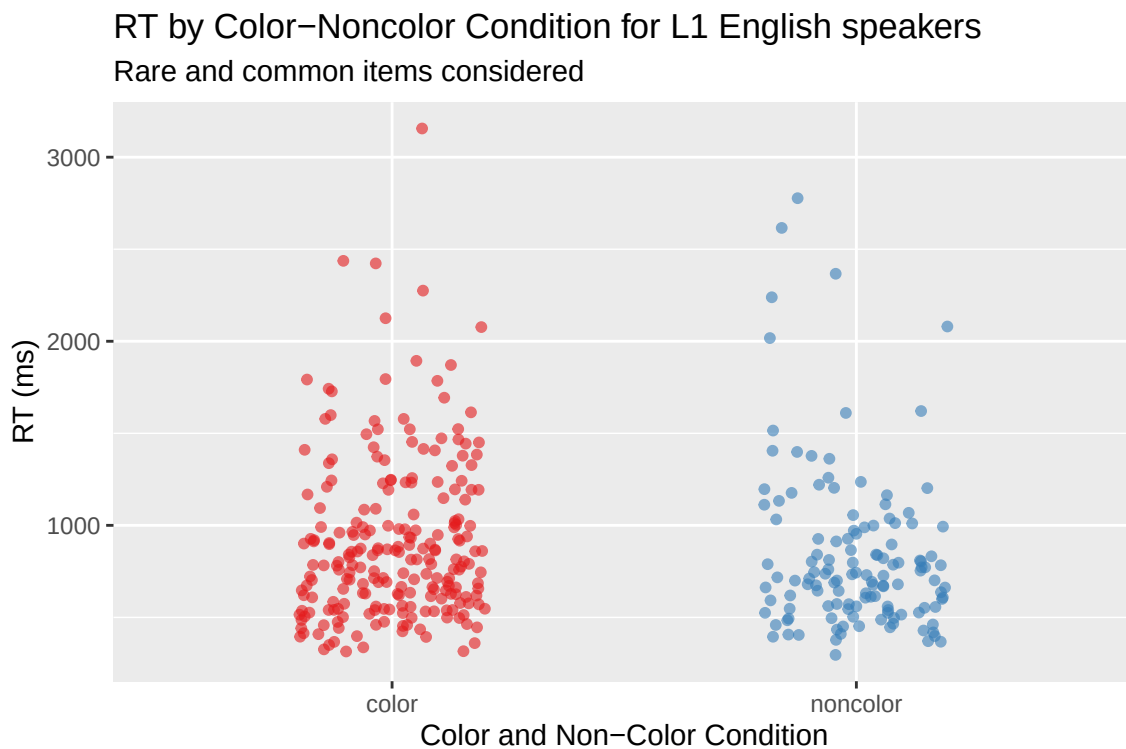
```
rawdata_L1_color_noncolor_condition <- rmdl_data_RT %>%
  filter(group == "English") %>%
  ggplot(aes(x = color_noncolor, y = RT, color=color_noncolor)) +
  geom_jitter(width = 0.2, alpha = 0.6, size = 1.4) +
  scale_color_brewer(
    name = "Color-Noncolor Condition",
```

```

labels = c("Color", "Noncolor"),
palette = "Set1",
type = "div",
guide="none") +
labs(
  x = "Color and Non-Color Condition",
  y = "RT (ms)",
  title = "RT by Color-Noncolor Condition for L1 English speakers",
  subtitle="Rare and common items considered")

rawdata_L1_color_noncolor_condition

```



```

ggsave("figures/rawdata_L1_color_noncolor_condition.svg")

```

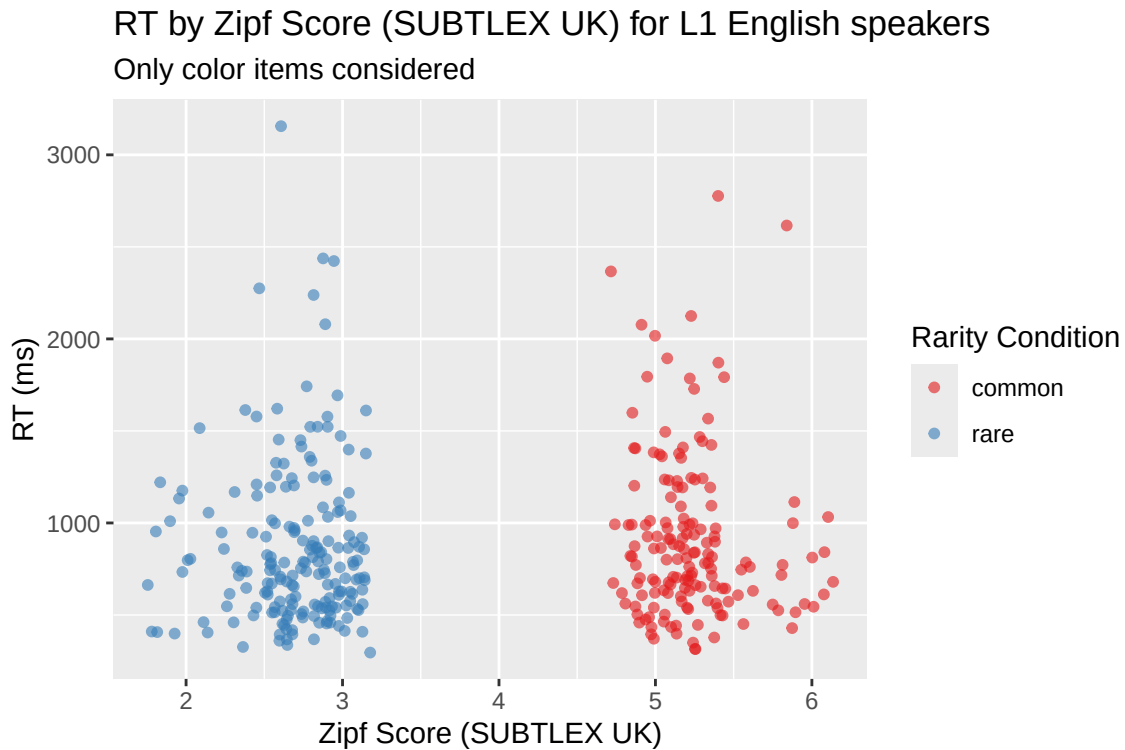
Saving 6 x 4 in image

Figure 2. This figure plots reaction times on the Y axis and the color-noncolor condition on the X axis. Only L1 English speakers are included in the plot, as this is a test of Hypothesis 1a; if L1 speakers of English are slower to respond to color words versus non-color words, there is

evidence of replicating the Stroop effect. For visual clarity, color word items are shown in red and non-color word items are shown in blue. This plot indicates that L1 English speakers in this study may have had slightly slower reaction times to color words as compared to non-color words, but the robustness of this difference will be assessed in the statistical model.

3.5.3 Figure 3.

```
rawdata_L1_color_condition <- rmdl_data_RT %>%  
  filter(group == "English") %>%  
  ggplot(aes(x = Zipf.SUBTLEX_UK, y = RT, color=rarity)) +  
  geom_jitter(width = 0.2, alpha = 0.6, size = 1.4) +  
  scale_color_brewer(  
    name = "Rarity Condition",  
    labels = c("common", "rare"),  
    palette = "Set1", type = "div") +  
  labs(  
    x = "Zipf Score (SUBTLEX UK)",  
    y = "RT (ms)",  
    title = "RT by Zipf Score (SUBTLEX UK) for L1 English speakers",  
    subtitle="Only color items considered")  
  
rawdata_L1_color_condition
```



```
ggsave("figures/rawdata_L1_color_condition.svg")
```

Saving 6 x 4 in image

Figure 3. This figure plots reaction times on the Y axis and Zipf score (SUBTLEX UK) as a correlate of rarity on the X axis. Only L1 English speakers are included in the plot, as this is a test of Hypothesis 1b; if L1 speakers of English have similar lexical entrenchment of rare and common color words, reaction times to both conditions should be similar. For visual clarity, common items are shown in red and rare items are shown in blue. This plot indicates that there may not be significantly different reaction times between common and rare colors among L1 speakers of English in this study.

3.6 Summarize the data

```
rmdl_data_RT %>%
  group_by(group) %>%
  summarise(
```

```

mean = mean(RT), median(RT), sd(RT), min(RT), max(RT), n = n()
) %>%
knitr::kable(
  digits = 2,
  col.names =
    c("Group", "Mean RT", "Median RT", "RT SD", "Min RT", "Max RT", "n")
) %>% kable_styling(full_width = F)

```

Group	Mean RT	Median RT	RT SD	Min RT	Max RT	n
English	886.09	778.5	442.74	296	3156	352
Mandarin	1048.89	890.0	507.88	469	4877	264

The data summary shows that L2 English (Mandarin) participants had higher reaction times on average, displaying a mean reaction time of 1048 ms compared to 886 ms for L1 English participants. L2 English participants also showed a median reaction time over 100 ms longer than L1 English participants. The standard deviation measures also show a slightly larger spread of L2 English responses as compared to L1 English responses. For L2 English participants, RT minima and maxima were both larger than L1 English participants.

4 Regression modelling

To model this data, I chose a LogNormal family. Reaction time data is often right-skewed, meaning it has a long tail to the right. Modeling with a LogNormal family “squishes” the data, allowing the model to interpret it as if it were close to a Gaussian distribution.

As for model parameters, this model explicitly seeks to understand the interaction between language group (L1 versus L2 English), stimulus item rarity (common versus rare), and color versus non-color word. This addresses the core hypotheses of the study, assessing L1 and L2 interactions with word frequency (rarity) and status as a color word. The model includes word length in letters to control for the reading effect that may be incurred from longer or shorter word length.

The model employs a maximal random effects model, including random slopes for each participant for word rarity, color-noncolor status, and word length. These random slopes were chosen as these were the three within-participant conditions that all participants experienced at different levels.

```

my_seed = 4423

rmdl_data_bm <- brm(
  RT ~ group * rarity * color_noncolor + Length +
    (rarity + color_noncolor + Length | participant_num),
  data = rmdl_data_RT,
  family = lognormal,
  iter = 4000,
  cores = 4,
  seed = my_seed,
  file = "cache/rmdl_bm",
  control = list(adapt_delta = 0.99, max_treedepth = 15)
)

```

Model Summary:

```
summary(rmdl_data_bm, prob = 0.95)
```

Family: lognormal
 Links: mu = identity
 Formula: RT ~ group * rarity * color_noncolor + Length + (rarity + color_noncolor + Length |
 Data: rmdl_data_RT (Number of observations: 616)
 Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 8000

Multilevel Hyperparameters:

~participant_num (Number of levels: 7)

	Estimate	Est.Error	1-95% CI	u-95% CI
sd(Intercept)	0.42	0.25	0.09	1.02
sd(rarityrare)	0.06	0.06	0.00	0.22
sd(color_noncolornoncolor)	0.08	0.08	0.00	0.27
sd(Length)	0.03	0.03	0.00	0.11
cor(Intercept,rarityrare)	-0.01	0.44	-0.81	0.81
cor(Intercept,color_noncolornoncolor)	-0.08	0.42	-0.82	0.73
cor(rarityrare,color_noncolornoncolor)	0.10	0.45	-0.77	0.86
cor(Intercept,Length)	-0.05	0.45	-0.84	0.80
cor(rarityrare,Length)	-0.01	0.46	-0.83	0.82
cor(color_noncolornoncolor,Length)	0.02	0.45	-0.81	0.82

	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.00	2473	2249
sd(rarityrare)	1.00	3106	3998

sd(color_noncolornoncolor)	1.00	2484	2668
sd(Length)	1.00	3145	3274
cor(Intercept,rarityrare)	1.00	7438	4890
cor(Intercept,color_noncolornoncolor)	1.00	8150	5765
cor(rarityrare,color_noncolornoncolor)	1.00	5082	5537
cor(Intercept,Length)	1.00	8779	4999
cor(rarityrare,Length)	1.00	7085	6281
cor(color_noncolornoncolor,Length)	1.00	6627	6552

Regression Coefficients:

	Estimate	Est.Error	l-95% CI		
Intercept	6.79	0.27	6.25		
groupMandarin	0.12	0.36	-0.61		
rarityrare	-0.09	0.06	-0.22		
color_noncolornoncolor	-0.14	0.08	-0.30		
Length	-0.01	0.03	-0.06		
groupMandarin:rarityrare	0.10	0.10	-0.09		
groupMandarin:color_noncolornoncolor	0.06	0.12	-0.16		
rarityrare:color_noncolornoncolor	0.08	0.07	-0.07		
groupMandarin:rarityrare:color_noncolornoncolor	0.07	0.11	-0.16		
	u-95% CI	Rhat	Bulk_ESS	Tail_ESS	
Intercept	7.33	1.00	3328	3320	
groupMandarin	0.87	1.00	3170	3123	
rarityrare	0.03	1.00	4256	3940	
color_noncolornoncolor	0.02	1.00	3483	3706	
Length	0.05	1.00	6970	5471	
groupMandarin:rarityrare	0.28	1.00	4097	4086	
groupMandarin:color_noncolornoncolor	0.30	1.00	3346	3306	
rarityrare:color_noncolornoncolor	0.22	1.00	5428	6056	
groupMandarin:rarityrare:color_noncolornoncolor	0.29	1.00	5262	6287	

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.34	0.01	0.32	0.36	1.00	12332	6122

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

4.1 Looking at MCMC draws from the model

```
rmdl_data_bm_draws <- as_draws_df(rmdl_data_bm)

# Now, following the QML textbook lesson 35,
#we transform the predictors relative relationships into combined values
rmdl_data_bm_draws_combined <- rmdl_data_bm_draws %>% mutate(
  eng_common_color = b_Intercept,
  mand_common_color = b_Intercept + b_groupMandarin,
  eng_rare_color = b_Intercept + b_rarityrare,
  eng_common_noncolor = b_Intercept + b_color_noncolornoncolor,
  mand_rare_color = b_Intercept + b_groupMandarin +
    b_rarityrare + `b_groupMandarin:rarityrare`,
  mand_common_noncolor = b_Intercept + b_groupMandarin +
    b_color_noncolornoncolor + `b_groupMandarin:color_noncolornoncolor`,
  eng_rare_noncolor = b_Intercept + b_rarityrare +
    `b_rarityrare:color_noncolornoncolor`,
  mand_rare_noncolor = b_Intercept + b_groupMandarin + b_rarityrare +
    b_color_noncolornoncolor +
    `b_groupMandarin:rarityrare` +
    `b_rarityrare:color_noncolornoncolor` +
    `b_groupMandarin:color_noncolornoncolor` +
    `b_groupMandarin:rarityrare:color_noncolornoncolor`,
) %>%
  select(eng_common_color, mand_common_color, eng_rare_color,
    eng_common_noncolor, mand_rare_color, mand_common_noncolor,
    eng_rare_noncolor, mand_rare_noncolor)
```

Warning: Dropping 'draws_df' class as required metadata was removed.

Now, we can look at the relevant variables described in Methods to address the research hypothesis.

4.1.1 Figure 4. L1 Stroop Replication

For hypothesis 1a, L1 reaction times in the color condition should be credibly slower than in the noncolor condition, regardless of rarity.

I am using the QML textbook Chapter 35 instructions to create a plot to assess the difference between these conditions.

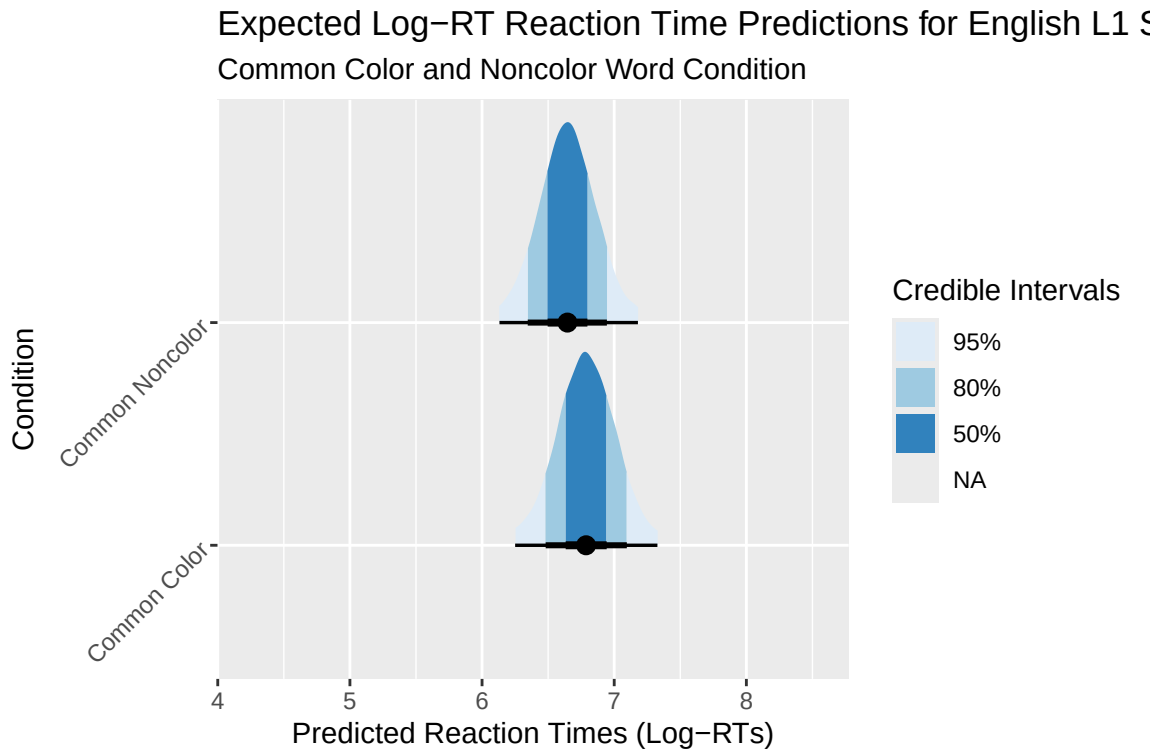
```

draws_combined_epred_H1 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, eng_common_noncolor) %>%
  pivot_longer(
    eng_common_color:eng_common_noncolor,
    names_to = "cond",
    values_to = "epred") %>%
  mutate(
    cond = case_when(
      cond == "eng_common_color" ~ "Common Color",
      cond == "eng_common_noncolor" ~ "Common Noncolor"
    )
  )

H1_epreds_plot <- draws_combined_epred_H1 %>%
  ggplot(aes(epred, cond)) +
  stat_halfeye(
    .width = c(0.5, 0.8, 0.95),
    aes(fill = after_stat(level))
  ) +
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  scale_fill_brewer(
    name="Credible Intervals",
    labels = c("95%", "80%", "50%")) +
  labs(
    title="Expected Log-RT Reaction Time Predictions for English L1 Speakers",
    subtitle="Common Color and Noncolor Word Condition",
    x="Predicted Reaction Times (Log-RTs)",
    y="Condition")

H1_epreds_plot

```



```
ggsave("figures/H1_epreds_plot.svg")
```

Saving 6 x 4 in image

4.1.2 Figure 5. L1 Rare and Common Colors - Reaction Time Equivalence Hypothesis

For hypothesis 1b, L1 reaction times should not be credibly different between the rare and common color conditions.

```
draws_combined_epred_H2 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, eng_rare_color) %>%
  pivot_longer(
    eng_common_color:eng_rare_color,
    names_to = "cond",
    values_to = "epred") %>%
  mutate(
    cond = case_when(
      cond == "eng_common_color" ~ "Common Color",
      cond == "eng_rare_color" ~ "Rare Color",
```

```

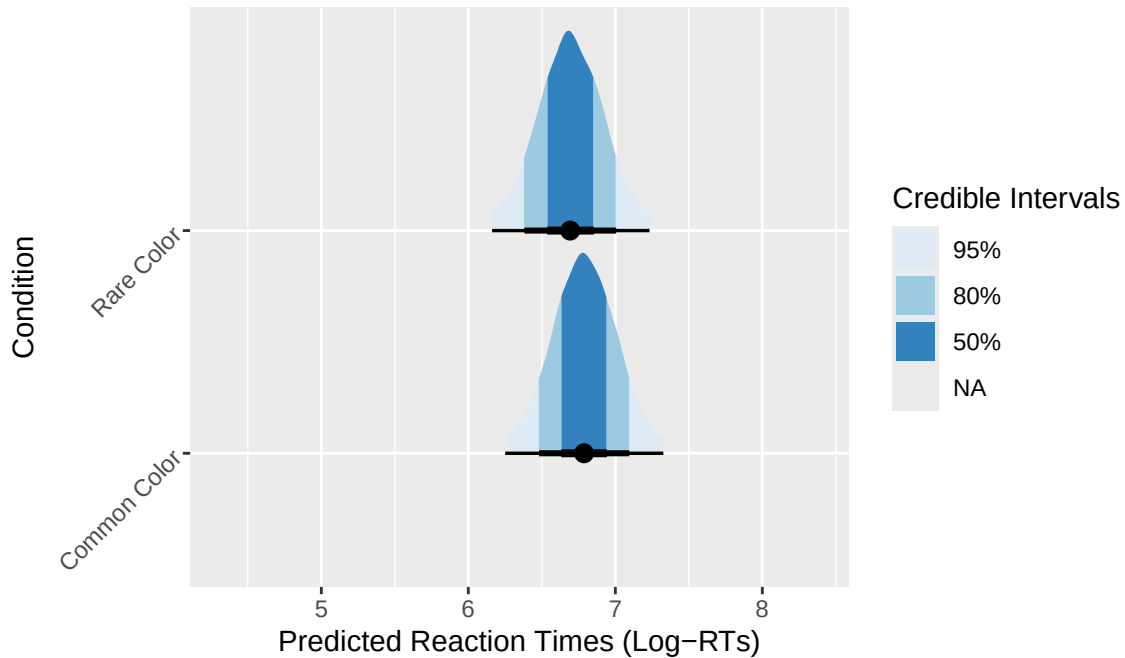
    TRUE ~ "" #Catch-all, should not ever be used if everything is working
  )
)

H2_epreds_plot <- draws_combined_epred_H2 %>%
  ggplot(aes(epred, cond)) +
  stat_halfeye(
    .width = c(0.5, 0.8, 0.95),
    aes(fill = after_stat(level))
  ) +
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  scale_fill_brewer(
    name="Credible Intervals",
    labels = c("95%", "80%", "50%")) +
  labs(
    title="Expected Log-RT Reaction Time Predictions for English L1 Speakers",
    subtitle="Between Rare and Common Color Words",
    x="Predicted Reaction Times (Log-RTs)",
    y="Condition")

H2_epreds_plot

```

Expected Log-RT Reaction Time Predictions for English L1 Sp Between Rare and Common Color Words



```
ggsave("figures/H2_epreds_plot.svg")
```

Saving 6 x 4 in image

4.1.3 Figure 6. L2 Faster Overall Reaction Times to Colors

For Hypothesis 2a, I assess the expected predictions for all L2 color responses versus all L1 color responses.

```
draws_combined_epred_H3 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, mand_common_color, eng_rare_color, mand_rare_color) %>%
  #Combining all color responses by averaging
  #while keeping L1 and L2 separate to compare color responses between groups
  mutate(
    L1_English = (eng_common_color + eng_rare_color) / 2,
    L2_English = (mand_common_color + mand_rare_color) / 2
  ) %>%
  pivot_longer(
    L1_English:L2_English,
```

```

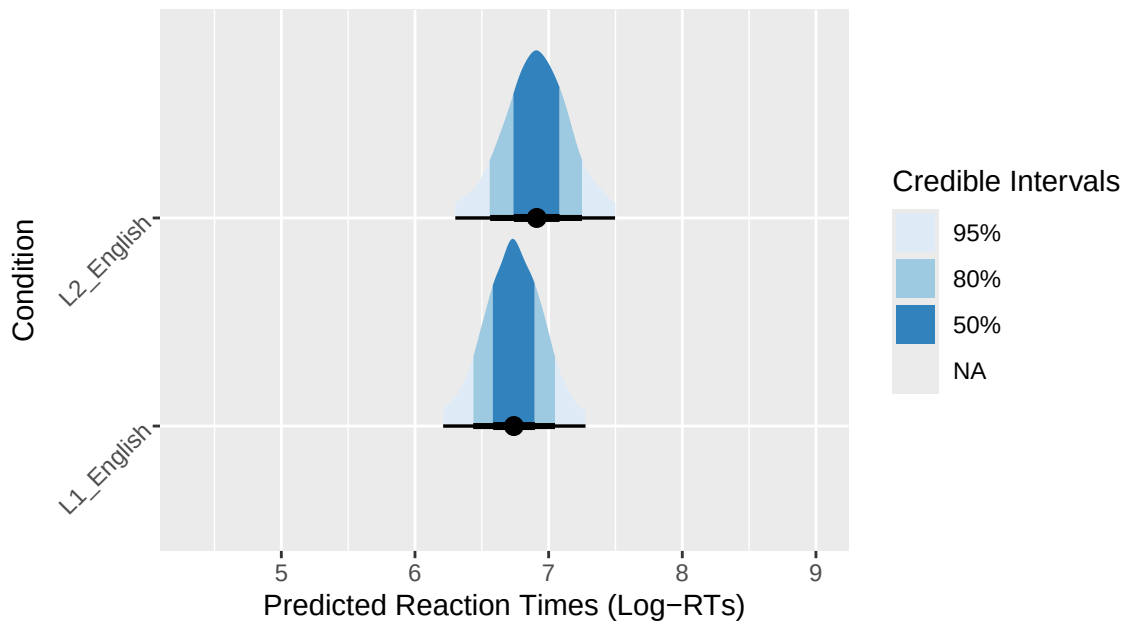
names_to = "cond",
values_to = "epred")

H3_epreds_plot <- draws_combined_epred_H3 %>%
  ggplot(aes(epred, cond)) +
  stat_halfeye(
    .width = c(0.5, 0.8, 0.95),
    aes(fill = after_stat(level))
  ) +
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  scale_fill_brewer(
    name="Credible Intervals",
    labels = c("95%", "80%", "50%")) +
  labs(
    title="Expected Log-RT Reaction Time Predictions
for English L1 and L2 Speakers",
    subtitle="Including Rare and Common Color Words",
    x="Predicted Reaction Times (Log-RTs)",
    y="Condition")

H3_epreds_plot

```

Expected Log-RT Reaction Time Predictions for English L1 and L2 Speakers Including Rare and Common Color Words



```
ggsave("figures/H3_epreds_plot.svg")
```

Saving 6 x 4 in image

4.1.4 Figure 7. L2 Greater Difference Between Rare and Common Color Reaction Times

Now we look at the statistical average differences between rare and common color RTs for L2 and L1 English speakers.

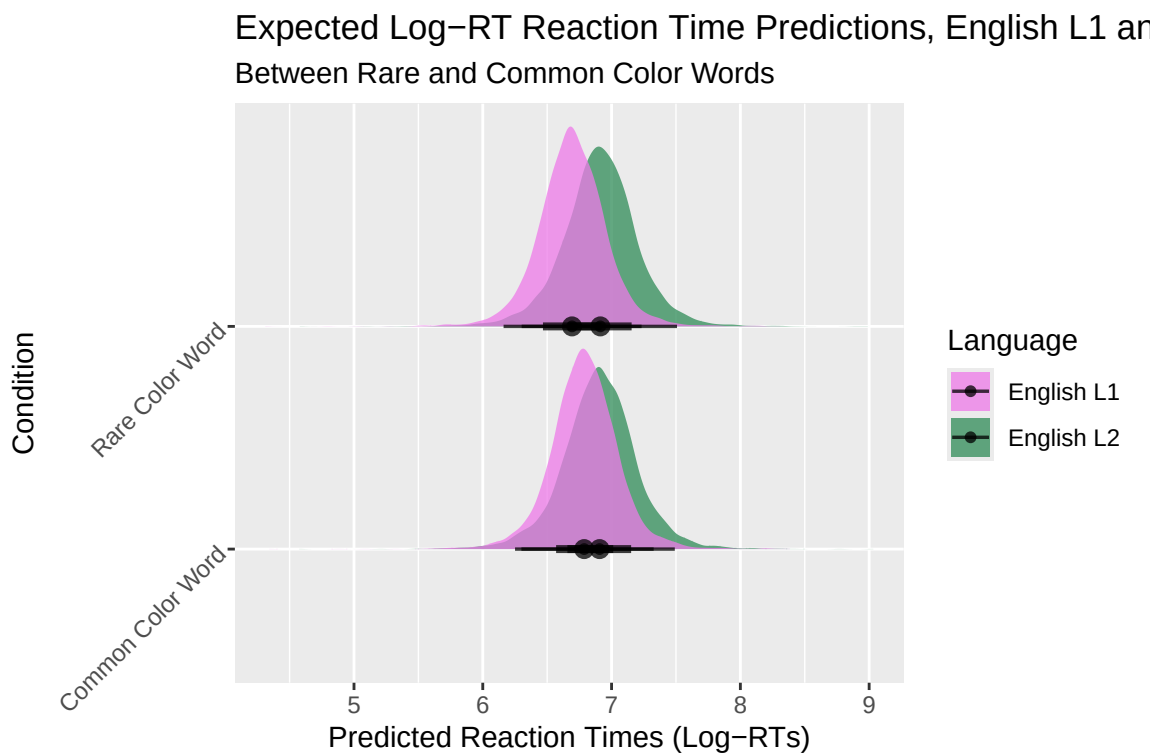
```
draws_combined_epred_H4 <- rmdl_data_bm_draws_combined %>%
  select(eng_common_color, mand_common_color, eng_rare_color, mand_rare_color) %>%
  pivot_longer(
    eng_common_color:mand_rare_color,
    names_to = "cond",
    values_to = "epred") %>%
  separate(cond, c("Language", "Rarity", extra = "merge")) %>%
  #Simple mutate to rename for plot Y axis labels
  mutate(
    Rarity = case_when(
```

```

    Rarity == "rare" ~ "Rare Color Word",
    Rarity == "common" ~ "Common Color Word",
    TRUE ~ "" #Catch-all, should not ever be used if everything is working
  )
)

draws_combined_epred_H4 %>%
  ggplot(aes(epred, Rarity, fill=Language)) +
  stat_halfeye(alpha=0.75) +
  scale_fill_manual(
    labels = c("English L1", "English L2"),
    values = c("orchid2", "seagreen")) +
  theme(axis.text.y = element_text(angle = 45, hjust = 1)) +
  labs(
    title="Expected Log-RT Reaction Time Predictions, English L1 and L2 Speakers",
    subtitle="Between Rare and Common Color Words",
    x="Predicted Reaction Times (Log-RTs)",
    y="Condition")

```



```
ggsave("figures/draws_combined_epred_H4.svg")
```

Saving 6 x 4 in image

5 Discussion

As shown by the model summary, the 95% credible interval for all predictors crossed 0. In other words, at 95% credibility, the model had no confidence in effect magnitude or direction.

These findings bear upon the hypotheses laid out in the Methods section of the report.

5.1 Hypothesis 1

The results of this study do not replicate the Stroop effect. At 95% credibility, there was no statistical difference in L1 English reaction times between color and non-color stimulus items in naming the stimulus font color.

Even so, L1 English participants did not have credibly longer or shorter reaction times between the common and rare color stimulus items in the study. While this finding aligns with the hypothesis laid out in the Methods section above, this result should be treated with caution; given that the study failed to replicate the Stroop effect, this result does not provide insight into automatic semantic inhibition based on word frequency among L1 speakers of English.

5.2 Hypothesis 2

As shown by the raw data summary, the prediction of hypothesis 2a that L2 English speakers would respond to all color words more quickly than L1 English speakers was not borne out by the results of this study. On average, L2 English responses were globally slower across all conditions. However, the statistical model did not find this difference credible at the 95% CrI, and therefore, there is no evidence from this study that L2 English speakers reliably respond differently than L1 English speakers to color words in the Stroop task paradigm.

Additionally, the statistical model's prediction for the interaction between L2 English speakers and word frequency (rarity) also crossed 0. This means that, at 95% credibility, we are not confident that high and low frequency English words change L2 English speakers' reaction times any differently than for L1 English speakers. This lack of confidence is shown clearly in Figure 8, where Log-RTs for both language groups overlap heavily.

Among the hypothesis set out in this study, the evidence does not credibly bear on any of them. Further studies should be conducted with more participants to better understand the relationship between the Stroop effect and word frequency. While past studies have argued

for a robust relationship between lexical decision tasks and word frequency in both L1 and L2 English populations, based on our results, we cannot make any claims as to the relationship between incongruous word-color naming and word frequency in the Stroop paradigm in either population.

5.3 Limitations

For the rare color word condition, this study used four rare color words unrelated to the font colors employed in the experiment. This is a severe limitation of the study, as it means that a congruous condition in rare color words is never observed by participants. Future studies should include length-matched rare and common words referring to the same font color to better explore this research question.

6 Bibliography

- Barr, Dale, Roger Levy, Christoph Scheepers, and Harry Tily. 2013. “Random Effects Structure for Confirmatory Hypothesis Testing: Keep It Maximal.” *Journal of Memory and Language* 68 (3): 255–78. <https://doi.org/10.1016/j.jml.2012.11.001>.
- Brysbaert, Marc, Evelyne Lagrou, and Michaël Stevens. 2017. “Visual Word Recognition in a Second Language: A Test of the Lexical Entrenchment Hypothesis with Lexical Decision Times.” *Bilingualism: Language and Cognition* 20 (3): 530–48. <https://doi.org/10.1017/S1366728916000353>.
- Diependaele, Kevin, Kristin Lemhöfer, and Marc Brysbaert. 2013. “The Word Frequency Effect in First- and Second-Language Word Recognition: A Lexical Entrenchment Account.” *Quarterly Journal of Experimental Psychology* 66 (5): 843–63. <https://doi.org/10.1080/17470218.2012.720994>.
- Scarpina, Federica, and Sofia Tagini. 2017. “The Stroop Color and Word Test.” *Frontiers in Psychology* 8 (April): 557. <https://doi.org/10.3389/fpsyg.2017.00557>.
- Stroop, John. 1935. “Studies of Interference in Serial Verbal Reactions.” *Journal of Experimental Psychology* (US) 18 (6): 643–62. <https://doi.org/10.1037/h0054651>.
- Taylor, Jack, Alistair Beith, and Sara Sereno. 2020. “LexOPS: An r Package and User Interface for the Controlled Generation of Word Stimuli.” *Behavior Research Methods* 52 (6): 2372–82. <https://doi.org/10.3758/s13428-020-01389-1>.

7 Appendix

7.1 A. Stimuli Generation and Table

The actual trial will reduce to 4 test colors to allow participants to lay 4 fingers over the 1-4 keyboard keys, increasing viability of reaction times. 8 keys may be too slow in baseline response to generate useful data.

with that in mind, let's cut the tibble down to size while keeping the word length balance.

```
lexops_df <- LexOPS::lexops

common_colors <- c("green", "brown", "blue", "black")
rare_colors <- c("sepia", "ochre", "mauve", "aqua")
common_noncolors <- c("fast", "slow", "big", "small")
rare_noncolors <- c("apex", "tusk", "nymph", "irk")

my_words <- c(common_colors, rare_colors, common_noncolors, rare_noncolors)

word_df <- tibble(string = my_words) %>%
  mutate(rarity = case_when(
    string %in% common_colors | string %in% common_noncolors ~ "common",
    string %in% rare_colors | string %in% rare_noncolors ~ "rare",
    TRUE ~ "other" # Catch-all, should never be used
  ))

word_df <- word_df %>%
  mutate(color_noncolor = case_when(
    string %in% common_colors | string %in% rare_colors ~ "color",
    string %in% common_noncolors | string %in% rare_noncolors ~ "noncolor",
    TRUE ~ "other" # Catch-all, should never be used
  ))

lookup_table <- lexops_df %>%
  select(string, Zipf.SUBTLEX_UK, Length)

word_df_final <- word_df %>%
  left_join(lookup_table, by = "string")

word_df_final <- word_df_final |>
  mutate(Zipf.SUBTLEX_UK = round(Zipf.SUBTLEX_UK, digits=2)) |>
  mutate(combined_condition = paste(rarity, color_noncolor, sep = "_"))
```

```
word_df_final %>%
  select(string, rarity, color_noncolor, Zipf.SUBTLEX_UK, Length) %>%
  knitr::kable(
    digits = 2,
    col.names = c("Word", "Rarity", "Color or Noncolor",
                  "SUBTLEX Zipf Score (UK)", "Length in Letters")
  ) %>% kable_styling(full_width = F)
```

Word	Rarity	Color or Noncolor	SUBTLEX Zipf Score (UK)	Length in Letters
green	common	color	5.22	5
brown	common	color	5.01	5
blue	common	color	5.16	4
black	common	color	5.26	5
sepia	rare	color	2.38	5
ochre	rare	color	2.77	5
mauve	rare	color	2.71	5
aqua	rare	color	2.98	4
fast	common	noncolor	5.09	4
slow	common	noncolor	4.81	4
big	common	noncolor	5.94	3
small	common	noncolor	5.43	5
apex	rare	noncolor	2.96	4
tusk	rare	noncolor	3.00	4
nymph	rare	noncolor	2.72	5
irk	rare	noncolor	1.95	3

```
write_csv(word_df_final, "testable_materials/exp_stimuli.csv")
```

7.2 B. Participant Language Proficiency Survey Results

```
survey_responses %>%
  knitr::kable(
    digits = 2,
    col.names = c("Participant ID", "English Proficiency",
                  "Mandarin Proficiency", "Other Languages")
  ) %>% kable_styling(full_width = F)
```

Participant ID	English Proficiency	Mandarin Proficiency	Other Languages
1	Native Speaker	No proficiency	NA
2	Native Speaker	No proficiency	NA
3	Native Speaker	No proficiency	NA
4	High Proficiency (CEFR C1-C2)	Native Speaker	NA
5	Intermediate Proficiency (CEFR B1-B2)	Native Speaker	NA
6	Native Speaker	No proficiency	NA
7	High Proficiency (CEFR C1-C2)	Native Speaker	NA

survey_responses

```
# A tibble: 7 x 4
  participant_num english_level      mandarin_level other_level
    <chr>          <chr>          <chr>          <lgl>
1 1              Native Speaker      No proficiency NA
2 2              Native Speaker      No proficiency NA
3 3              Native Speaker      No proficiency NA
4 4              High Proficiency (CEFR C1-C2) Native Speaker NA
5 5              Intermediate Proficiency (CEFR B1-B2) Native Speaker NA
6 6              Native Speaker      No proficiency NA
7 7              High Proficiency (CEFR C1-C2) Native Speaker NA
```

7.3 C. Building the Testable file

We then create a “key” tibble to map keypress responses to font color.

```
keypress_key <- tibble(
  color = c("stim1: black", "stim1: blue", "stim1: green", "stim1: brown"),
  key = c(1, 2, 3, 4)
)
```

We cross the conditions to get the experimental stimuli and keys.

```
exp_trialfile <- crossing(word_df_final, keypress_key)
```

Now, let’s finalize the Testable trial file to label conditions as “congruent” (text color name is the same as font color), “incongruent” (text color name is NOT the same as font color), or “control” (text color name cannot be related to font color).

```
exp_trialfile <- exp_trialfile %>% mutate(
  congruence_condition = case_when(
    color_noncolor == "color"
    & rarity == "common" & paste0("stim1: ", string) == color ~ "congruent",
    color_noncolor == "color"
    & rarity == "common" & paste0("stim1: ", string) != color ~ "incongruent",
    TRUE ~ "control"
  )
)
```

We want to get a 1:2 ratio of critical vs control trials

```
exp_trialfile_crit <- exp_trialfile %>%
  filter(color_noncolor == "color") %>%
  crossing(1:2)

exp_trialfile_cont <- exp_trialfile %>%
  filter(color_noncolor != "color")

exp_trialfile <- bind_rows(exp_trialfile_crit, exp_trialfile_cont)
```

Let's add in the instructions and practice trials for testable.

```
head_block1 <- tibble(
  type = "instructions",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI = 0,
  button1 = "NEXT",
  keyboard = "",
  random = 1,
  required = "",
  title = "Welcome to the experiment",
  content = "First, we'd like to collect some information about your language"
```

```

    background. Please click NEXT and answer the questions on the form.",
    head = "",
    responseType = ""
)

#English Proficiency Check Row
head_block2 <- tibble(
  type = "form",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI = 0,
  button1 = "NEXT",
  keyboard = "",
  random = 2,
  title = "",
  content = "",
  responseType = "radio",
  required = "1",
  head = "What is your English language READING ability?",
  responseOptions = "Native Speaker;
High Proficiency (CEFR C1-C2);
Intermediate Proficiency (CEFR B1-B2);
Low Proficiency (CEFR A1-A2);
No proficiency:exclusive"
)

#Mandarin Proficiency Check Row
head_block3 <- tibble(
  type = "form",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",

```

```

stimFormat = "",
ITI      = 0,
button1 = "",
keyboard = "",
random = 3,
title = "",
content = "",
responseType = "radio",
required = "1",
head = "What is your Standard Mandarin ( ) language READING ability?",
responseOptions = "Native Speaker;
High Proficiency (CEFR C1-C2);
Intermediate Proficiency (CEFR B1-B2);
Low Proficiency (CEFR A1-A2);\
No proficiency:exclusive"
)

head_block4 <- tibble(
  type = "form",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI      = 0,
  button1 = "",
  keyboard = "",
  random = 4,
  title = "",
  content = "",
  responseType = "box",
  required = "",
  head = "Are you proficient in or learning any other foreign languages?
(Please include the name of the language and your proficiency self-assessment.)",
  responseOptions = ""
)

head_block5 <- tibble(
  type = "instructions",

```

```

stim1 = "",
color = "",
key = 0,
condition1 = "",
condition2 = "",
feedback = "",
trialText = "",
stimFormat = "",
ITI = 0,
button1 = "NEXT",
keyboard = "",
random = 5,
required = "",
title = "Thank you for your survey responses! ",
content = "Let's move on to the experiment.
\nIn this study, you are asked to indicate the FONT COLOR of the text,
ignoring the meaning of the word. Please use the keyboard keys 1, 2, 3, and 4
to indicate the FONT COLOR of the word presented on the screen:<br><br>
Press 1 for BLACK, <br>2 for <span style=\"color:blue;\">BLUE</span>,
<br>3 for <span style=\"color:green;\">GREEN</span>, and <br>4 for
<span style=\"color:brown;\">BROWN</span> <br><br>
Please answer as accurately and as quickly as possible.
Let's do some practice!",
head = "",
responseType = ""
)

practice_block <- tribble(
  ~stim1, ~color, ~key,
  "BLACK", "stim1: black", 1,
  "GREEN", "stim1: black", 1,
  "BLUE", "stim1: blue", 2,
  "BROWN", "stim1: green", 3,
  "TUSK", "stim1: blue", 2,
  "SEPIA", "stim1: green", 3,
  "IRK", "stim1: brown", 4,
  "NYMPH", "stim1: brown", 4
) %>%
mutate(
  type = "test",
  condition1 = "practice",

```

```

feedback = glue::glue('correct: Correct! You responded in %RT% ms ;
                        incorrect: Incorrect! The font color was
                        {toupper(substring(color, 7))}.
                        You responded in %RT% ms'),

stimFormat = "word",
ITI = 400,
keyboard = "1 2 3 4",
random = 6,
required = "",
title = "",
content = "",
trialText = "Press 1 for BLACK, <br>2 for <span style=\"color:blue;\">BLUE
</span>, <br>3 for <span style=\"color:green;\">GREEN</span>,
and <br>4 for <span style=\"color:brown;\">BROWN</span>",
button1 = "",
head = "",
responseType = ""
)

head_block6 <- tibble(
  type = "instructions",
  stim1 = "",
  color = "",
  key = 0,
  condition1 = "",
  condition2 = "",
  feedback = "",
  trialText = "",
  stimFormat = "",
  ITI = 0,
  button1 = "NEXT",
  keyboard = "",
  random = 7,
  required = "",
  title = "Ready to start?",
  content = "Remember: Press 1 for BLACK, <br>2 for <span style=\"color:blue;\">BLUE
</span>, <br>3 for <span style=\"color:green;\">GREEN</span>,
and <br>4 for <span style=\"color:brown;\">BROWN</span>",
  head = "",
  responseType = ""
)

```

Now we transform it into a Testable-usable file:

```
exp_trialfile_main <- exp_trialfile %>% mutate(
  type = "test",
  stim1 = toupper(string),
  condition1 = case_when(
    congruence_condition != "control" ~ "critical",
    congruence_condition == "control" ~ "filler"
  ),
  condition2 = congruence_condition,
  trialText = case_when(
    type != "instructions" ~ "Press 1 for BLACK, <br>2 for
    <span style=\"color:blue;\">BLUE</span>, <br>3 for
    <span style=\"color:green;\">GREEN</span>,
    and <br>4 for <span style=\"color:brown;\">BROWN</span>",
    TRUE ~ ""
  ),
  feedback = "",
  stimFormat = case_when(
    type != "instructions" ~ "word",
    TRUE ~ ""
  ),
  ITI = case_when(
    type != "instructions" ~ 400,
    TRUE ~ 0
  ),
  button1 = case_when(
    type == "instructions" ~ "NEXT",
    TRUE ~ ""
  ),
  keyboard = case_when(
    type != "instructions" ~ "1 2 3 4",
    TRUE ~ ""
  ),
  random = 8,
  required = "",
  title = "",
  content = "",
  head = "",
  responseType = ""
)
```

Now let's save it for use in Testable.

```
exp_trialfile_final <- bind_rows(  
  head_block1, head_block2, head_block3, head_block4, head_block5,  
  practice_block, head_block6, exp_trialfile_main) %>%  
  mutate(stimSize = "175%")  
  #This just makes the main word on the screen bigger for participants.  
  
write_csv(exp_trialfile_final, "testable_materials/exp_trialfile.csv")
```